

Figure 1: Traffic captured using HUB

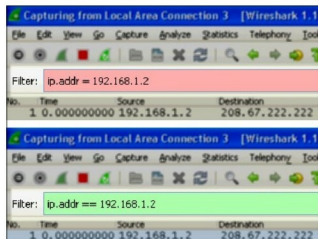


Figure 2: Default Wireshark display filters

feature called 'Display Filter' can be used very effectively.

Wireshark's Display Filter

This helps to sort/view the network traffic using various parameters such as the traffic originating from a particular IP or MAC address, traffic with a particular source or destination port, ARP traffic and so on. It is impossible to imagine Wireshark without display filters!

Click on 'Expressions' or go to 'Analyse – Display filters' to find a list of pre-defined filters available with Wireshark. You can create custom filters depending upon the analysis requirements—the syntax is really simple.

As seen in Figure 2, the background colours of the display filter box offer ready help while creating proper filters. A green background indicates the correct command or syntax, while a red background indicates an incorrect or incomplete command. Use these background colours to quickly identify syntax and gain confidence in creating the desired display filters.

A few simple filters are listed below:

- tcp*: Displays TCP traffic only
- arp*: Displays ARP traffic
- eth.addr == aa:bb:cc:dd:ee:ff*: Displays traffic where the Ethernet MAC address is aa:bb:cc:dd:ee:ff
- ip.src == 192.168.51.203*: Displays traffic where the source IP address is 192.168.51.203
- ip.dst == 4.2.2.1*: Displays traffic where the destination IP address is 4.2.2.1
- ip.addr == 192.168.51.1*: Displays traffic where the

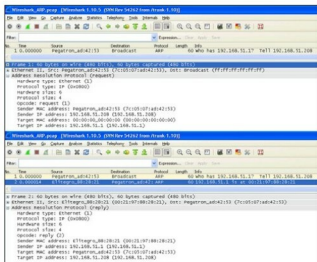


Figure 3: ARP protocol

source or the destination IP address is 192.168.51.1

Click on 'Save' to store the required filter for future use. By default, the top 10 custom filters created are available for ready use under the dropdown menu of the 'Filter' dialogue box.

With this background, let us look at two simple protocols—ARP and DHCP.

Address Resolution Protocol (ARP)

This is used to find the MAC address from the IP address. It works in two steps—the ARP request and ARP reply. Here are the details.

Apply the appropriate display filter (ARP) and view only ARP traffic from the complete capture. Also, refer to Figure 3 - the ARP protocol.

The protocol consists of the ARP request and ARP reply.

ARP request: This is used to find the MAC address of a system with a known IP address. For this, an ARP request is sent as a broadcast towards the MAC broadcast address:

Sender MAC address - 7c:05:07:ad:42:53
 Sender IP address - 192.168.51.208
 Target MAC address - 00:00:00:00:00:00
 Target IP address - 192.168.51.1



Note: Target IP address indicates the IP address for which the MAC address is requested.

Wireshark displays the ARP request under the 'Info' box as: *Who has 192.168.51.1? tell 192.168.51.208*

ARP reply: This ARP request broadcast is received by all systems connected to the network segment of the sender (below the router), mind well, this broadcast also reach router port connected to this segment.

The system with the destination IP address mentioned in the ARP request packet replies with its MAC address via an ARP reply. The important contents of the ARP reply are: